How-To Library

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Configuring the Messaging Mode in Vibe Data Stream for Machine Data

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Abstract

You can configure Vibe Data Stream data flows to use Streaming or Load Balancing messaging mode to publish data. This article describes the Streaming and Load Balancing messaging modes, guidelines to select the messaging mode, and the steps to configure the messaging mode in Informatica Vibe Data Stream for Machine Data 2.1.0.

Supported Versions

- Vibe Data Stream for Machine Data 2.1.0

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Messaging Mode Overview

Vibe Data Stream for Machine Data (VDS) does not stage data at any point between the data source and the target. To transport data, VDS uses the publish/subscribe model. VDS generates a topic name for a data flow. Source services read data in blocks and publish messages on a topic. Target services subscribe to that topic, and they receive those messages and write them to the target data store.

You can configure source services to use streaming or load balancing messaging mode to publish data.

Streaming

You can configure the VDS data flows to use the streaming messaging mode, called Ultra Messaging Streaming, to send data to targets across a network. When you configure the streaming mode, VDS delivers all the messages published by the source services to all the target services. As a result, the VDS Nodes associated with multiple target services get duplicate data.

You can configure the messaging mode when you create or edit a data flow. Before you change the messaging mode, you must undeploy a data flow. You can also specify advanced Ultra Messaging configuration for the source services, target services, and transformations in the data flow.

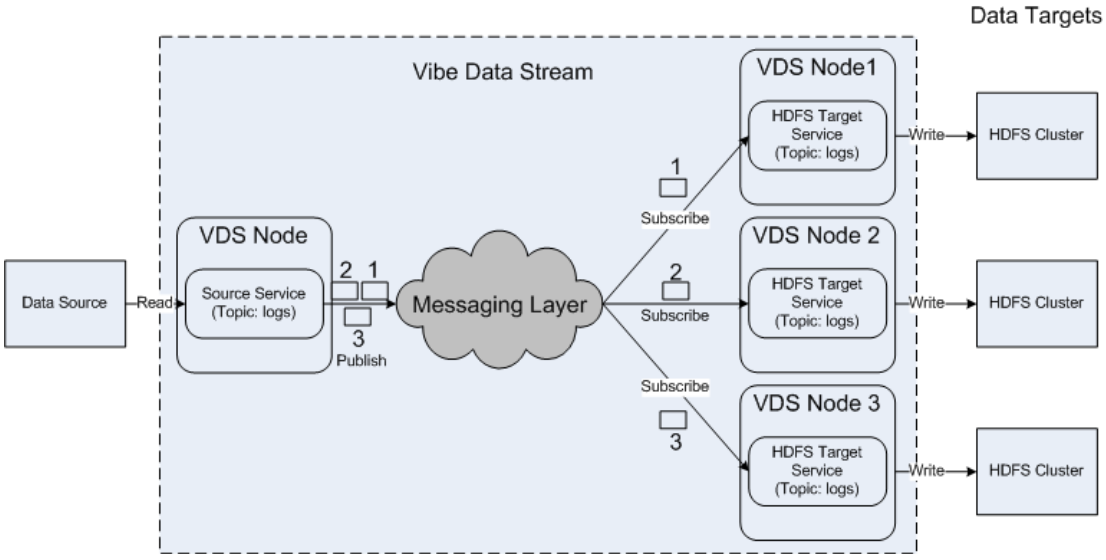
Load Balancing

You can configure the VDS data flows to use the Load Balancing message mode to reduce the load on a target service. To reduce the load on a target service, you can deploy the target service on multiple nodes, and you can configure the target service instances to write data to the target concurrently. VDS detects the presence of multiple instances of a target service and balances the load across the instances.

A target service sends a source service an acknowledgment after the target service processes a message. When a data flow contains multiple target services, a source service stops publishing messages when the number of unacknowledged messages reaches 1,000. A source service identifies a message as acknowledged only if it receives an acknowledgment from all the target services that receive the message.

When you deploy a target service on multiple VDS Nodes, VDS uses the round-robin method to distribute the messages across the instances. VDS avoids data duplication within the load balanced group of instances by delivering each message to one instance. Deploying a target service on multiple nodes ensures that a large volume of data from multiple sources does not overwhelm a single target service instance.

The following image shows how VDS balances the load across HDFS target services that you deployed on three nodes:



VDS balances the load as follows:

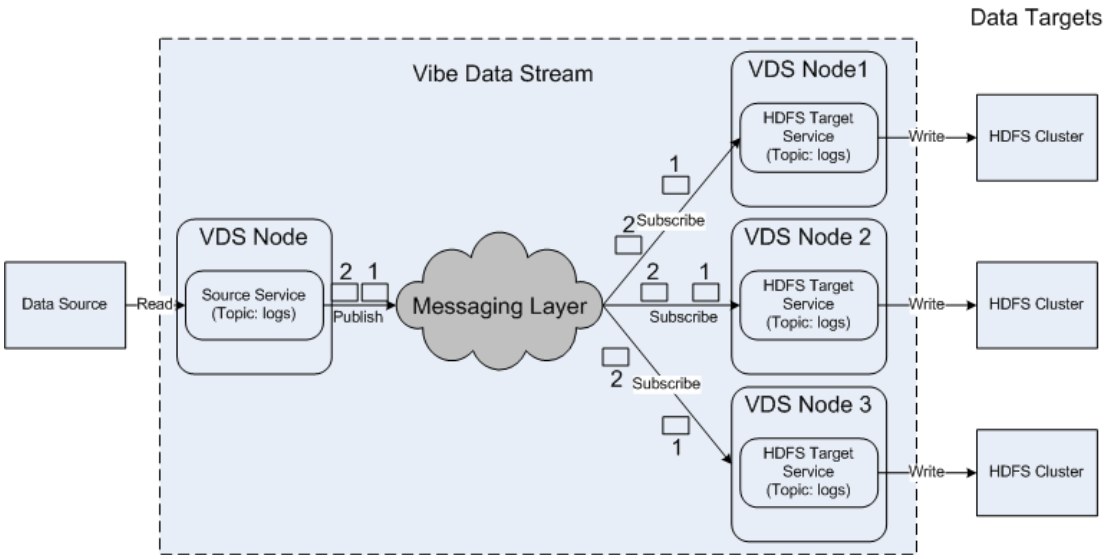
1. A source service reads data from the data source and publishes it as three messages over a topic called logs.
2. An HDFS target service instance runs on three nodes.
3. You have deployed the target service on three nodes for purposes of load balancing. VDS balances the load across the three instances of the target service in round-robin fashion.

Note: The distribution of messages depends on the load on individual instances. Although VDS uses the round-robin method, some instances might receive more messages than other instances.

Data Duplication

If you do not configure load balancing, VDS duplicates the data across the data targets. VDS delivers all the messages published by the source service to all the data target services so that they have the complete data set for analysis.

The following image shows VDS data duplication across the data targets:



VDS duplicates data as follows:

1. A source service reads data from the data source and publishes it as two messages over a topic called logs.
2. An HDFS target service instance runs on three nodes.
3. VDS delivers all the messages to all three target services.
4. The target services receive both messages and write them to the targets.

Guidelines for Selecting the Messaging Mode

Consider the following guidelines when you select the message mode:

- Use the Streaming messaging mode when you write from one source service to one target service.
- Use the Load Balancing message mode when you want to avoid data duplication.
- When you use HTTP and File source service types, use the Load Balancing messaging mode when you want synchronous delivery, that is, a response to be sent to the client when the message reaches the target service.

Configuring the Messaging Mode

Create a data flow and specify global data flow properties, such as the type of topic resolution that you want to use.

1. Log in to the Administrator tool.
2. On the **Domain** tab, click **Services and Nodes**.
3. In the **Domain Navigator** pane, click the Vibe Data Stream service.
4. Click the **Vibe Data Stream** tab.
5. In the **Data Flows** pane, right-click **All data flows**.
6. Click **New data flow**.

The **Data flow** dialog box appears.

7. In the **Name** field, enter a name for the data flow.
8. From the **Topic resolution type** list, select the type of topic resolution that you want to use.

Option	Description
Multicast	Publish messages over multicast.
Unicast	Publish messages over unicast.

9. In the **Resolver Address** field, enter the address of the topic resolution domain.

   If you choose unicast topic resolution, the resolver address is the address of the unicast resolution daemon. If you choose multicast topic resolution, the resolver address is the multicast address that you want to use.
10. From the Messaging Mode list, select the mode in which the source service distributes data to the target service.

Option	Description
Load Balancing	Enables the source service to use load balancing to distribute data.
Streaming	Enables the source service to use streaming to distribute data.

11. Click **Save**.

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